

Exercise Sheet

Task 1 – Replacement Strategies

In an associative cache with four cache lines, the following sequence is executed: 0xB43, 0x427, 0x67C, 0xC6A, 0xB43, 0x406, 0xDA7, 0xE59, 0x406, 0xB43, 0x5E4, 0x427, 0xE59, 0xC6A, 0x406, 0xE59. Decide the cache's content for each of the following replacement strategies.

- a) FIFO
- b) LFU with LRU Tie-Break
- c) Clairvoyant

If multiple addresses won't appear again in the future, choose the left-most line to replace its content

Task 2 – Set-Associative Caches

Access sequence: 0xABC, 0xB43, 0xC6A, 0xABC 0xDD4, 0xC6A 0xE2F, 0xABC

Execute the above sequence in a set-associative cache with 2 cache sets (Set 0 and Set 1) and 4 cache lines altogether and uses LRU as replacement policy. An address is put into a cache set that has the same number as its LSB. Decide for each access whether a cache hit or a cache miss happens. If a cache miss occurs, elaborate what type of miss happened.